

FOURA — AUTONOMOUS REVENUE RECOVERY

Razorpay AI Buildathon 2026 · Track 03: AI Revenue Recovery · fursatiinsaان

1. SYSTEM ARCHITECTURE & 3 CONSOLIDATED HUBS

Foura intercepts failed digital checkouts in real-time, diagnoses root causes via **LLaMA-3 on Groq**, verifies **deterministic safety guardrails**, and dispatches recovery actions in **<0.82 seconds**.

Hub Name	Operational Role	Key Capabilities
1. Recovery Hub	Real-Time Cockpit & Execution	4 KPI cards, USD/EUR/INR live toggle, synthetic checkout failure simulator, pending queue, LLaMA-3 inspector, WhatsApp concierge card, auto-pilot mode.
2. Transactions Hub	Immutable Audit Ledger	Searchable table, inline expandable Framer Motion forensic drawers, ISO 8583 error reasons, instant multi-currency settlement payouts, 1-click CSV export.
3. Engine Hub	Math & Compliance Control	Multi-Agent 4-Node DAG, UCB1 bandit policy formulas, live rail latency radar (Visa, SEPA, UPI), interactive GMV profit lift slider, RBI retry limits.

2. THE 5 FINTECH ENGINEERING WAR STORIES

War Story 1: The Asynchronous Ingestion Race Condition

When simulating failures or receiving webhooks, returning the `payment_id` before background tasks committed resulted in intermittent 404s on immediate recovery clicks. **Solution:** The Synchronous Ingestion Boundary commits a placeholder SQLite record before spawning the background worker, ensuring zero race conditions.

War Story 2: Sandwiched Determinism Against Financial Hallucinations

LLMs cannot be trusted with financial amounts or discounts. **Solution:** The AI Sandwich. Deterministic Python calculates all order values and currency conversions before the prompt; post-generation, a deterministic safety circuit clamps any discount to a strict 5% margin ceiling.

War Story 3: Hard Decline Credential Safety Gate

Mistyped UPI MPINs (`BAD_REQUEST_PAYMENT_PIN_INCORRECT`) or expired cards lock bank accounts if retried. **Solution:** The engine immediately halts retries, flags `HARD_DECLINE_SECURITY_BLOCKED`, assigns 1.0 confidence, and advises the user to contact their bank.

War Story 4: Self-Healing Virtualenv Interpreter Launcher

macOS Python 3.14 symlink drift broke virtualenvs on clean machines. **Solution:** `run.py` actively tests interpreter execution with subprocess and automatically self-heals and reinstalls dependencies if corrupted.

War Story 5: Mobile Layout Isolation & Clean Styling

Unclosed `@media` queries in CSS swallowed modal styling and caused responsive layout bugs. **Solution:** Isolated media queries, cleanly scoped modal dialogs, and added keyframe animations for zero-jitter rendering.

3. THE 5-MINUTE VIDEO DEMO SCRIPT (SECOND-BY-SECOND)

[0:00 - 0:45] The Hook & The Problem

Action: Display Recovery Hub on localhost:5173. Topbar telemetry ticker is pulsing.

"Hi everyone, I'm presenting Foura — an Autonomous AI Revenue Recovery Engine built for Track 03 of the Razorpay AI Buildathon. Across global e-commerce, over 20% of high-intent checkouts fail due to bank switch degradation, 3DS OTP challenge timeouts, or price hesitation. Traditional recovery relies on slow email reminders sent hours later, or dumb cron retries that trigger fraud blocks. Foura intercepts payment failures in real-time, diagnoses root causes using LLaMA-3 on Groq, verifies deterministic compliance guardrails, and recovers revenue in under 0.82 seconds."

[0:45 - 1:45] Live Demo: Interception & LLaMA-3 Diagnostics

Action: Select Cloud Infrastructure (\$1,299), select 3DS_OTP_CHALLENGE_TIMEOUT, click 'Simulate Failure & Auto-Recover'. Listen for Web Audio chime and watch confetti.

"Let's watch it live. A customer, Sarah Jenkins, tries to purchase a \$1,299 subscription, but her bank's 3DS SMS challenge times out. In under 12ms, Foura intercepts the drop. The 4-node execution pipeline kicks in: Node 02 diagnoses an involuntary technical drop via LLaMA-3; Node 03 verifies our RBI retry caps; Node 04 generates a signed recovery link. Sarah receives a tailored WhatsApp Concierge link reserving her cart for 15 minutes with 1-click payment."

[1:45 - 2:30] The Safety Circuit: Stopping Hard Declines

Action: Select BAD_REQUEST_PAYMENT_PIN_INCORRECT, click simulate, point to HARD_DECLINE_SECURITY_BLOCKED tag.

"Here is our key innovation: The Bounded AI Sandwich. When a customer enters an incorrect UPI PIN, standard bots would retry and lock the user's bank account. Foura's deterministic safety circuit intercepts this at the boundary, overrides the LLM, halts automated retries, and marks it HARD_DECLINE_SECURITY_BLOCKED. 100% compliant with RBI and global banking regulations."

[2:30 - 3:30] Multi-Currency & Immutable Transactions Hub

Action: Toggle currency to INR and EUR. Open Transactions & Payouts tab, click Forensics, click Instant Settle and Export CSV.

"Foura is multi-currency native. Switching to INR or EUR recalculates revenue exposure in real time. In the Transactions Hub, every row features an inline expandable forensic drawer with ISO error codes and LLaMA-3 diagnostics. Merchants can click 'Instant Settle' for on-demand same-day payouts across USD Fedwire, EUR SEPA, or IMPS, or export the full CSV ledger."

[3:30 - 4:15] Engine Architecture & Multi-Armed Bandit Math

Action: Open Engine Architecture tab, show 4-node DAG cards, point to UCB1 formula, slide GMV from \$250k to \$1M.

"In our Engine view, you can see our multi-agent pipeline and the mathematical brain powering policy selection: the Multi-Armed Bandit UCB1 algorithm. It balances exploitation of high-converting WhatsApp links with exploration of smart UPI rail switches. For a merchant doing \$1M monthly GMV, Foura recovers over \$69,000 every month — an \$800,000 annual profit expansion."

[4:15 - 5:00] Wrap-up & Production Readiness

Action: Show Swagger docs at /docs with Foura logo, run 'python3 tests/test_e2e_suite.py' in terminal showing 11/11 tests pass.

"To ensure enterprise reliability, we built an 11-point end-to-end automated test suite covering all endpoints, safety guardrails, and Razorpay webhook ingestion. The frontend builds in 125 milliseconds with zero linter warnings. Everything boots with a single command: python3 run.py. Thank you!"

4. JUDGE INTERVIEW Q&A; MASTER CHEAT SHEET

Q: Why not just use Razorpay webhooks with a cron job?

A: Cron jobs are dumb retries. If an issuer switch is flapping or the user entered the wrong PIN, a cron job blindly re-charges the card, triggering bank velocity lockouts and interchange fees. Foura uses LLaMA-3 to understand the root cause — distinguishing technical drops from price hesitation — and routes the customer to alternate rails with dynamic cart hold concessions.

Q: How do you prevent LLaMA-3 from hallucinating financial amounts?

A: Through the AI Sandwich Architecture. The LLM has zero authority over numbers. Order values, exchange rates, and discount caps are calculated deterministically in Python before the prompt is formatted. Post-generation, our safety circuit validates that any concession never breaches our 5% gross margin ceiling.

Q: What happens if the customer's internet goes down during checkout?

A: That drops as an ISO 8583 timeout (68 or 96). Foura intercepts it in under 12ms, generates an idempotent payment link via Razorpay SDK, and sends an automated WhatsApp Concierge message: 'We've saved your cart for 15 minutes — tap here to complete in one click.'

Q: What happens when a transaction hits 3 retries?

A: Under RBI and global card scheme regulations, repeated retries on a failing instrument risk merchant fines and card blacklisting. Foura strictly enforces an automated stopping rule: at retry count ≥ 3 , the transaction is permanently halted under RBI_3_RETRY_LIMIT_BREACH.

Q: How does the Multi-Armed Bandit (UCB1) formula work here?

A: Static A/B testing wastes revenue because 50% of traffic is sent to a losing strategy. UCB1 dynamically shifts traffic toward the highest-converting recovery channel (exploitation) while reserving a calculated percentage of attempts (exploration) to discover if new

rails perform better as bank switches fluctuate.